

CRISIS DISPATCHER

Behavioural Assessment Platform:

For Measuring Emotional Regulation Using an Emergency Response Simulation

Submitted by:

Evlin Sara Johny & Lakshmi Pramode

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Emotional regulation is the ability to manage emotions and make effective decisions during stressful situations. It is an essential psychological skill that influences problem-solving, adaptability, and resilience. Traditional assessment methods mainly rely on questionnaires or physiological sensors, which can be subjective or expensive.

This project proposes a software-based behavioural assessment platform that evaluates emotional regulation using an emergency response simulation. Participants complete realistic emergency scenarios while the system records behavioural data such as reaction time, decision accuracy, recovery time, and strategy changes. These metrics are analysed to generate an Emotional Regulation Score.

1.2 Problem Statement

Existing emotional regulation assessments are either subjective or require costly equipment. Most simulation systems focus on task performance rather than behavioural indicators of emotional regulation. Therefore, there is a need for an accessible, software-based platform that objectively measures emotional regulation through behavioural analytics.

1.3 Objectives

The objectives of this project are:

- * Develop an emergency response simulation for behavioural assessment.
- * Record user interactions during stressful situations.
- * Measure stress tolerance, recovery, persistence, and decision stability.
- * Generate an Emotional Regulation Score.
- * Provide behavioural analytics through an interactive dashboard.

1.4 Scope

The proposed system focuses on behavioural assessment using simulated emergency scenarios. It records user interactions, analyses behavioural metrics, and generates an assessment report. The MVP includes one complete assessment scenario with event logging, scoring, and analytics.

Target users = **Higher Secondary Students (Grades 11–12)**

1.5 Expected Outcome

The system is expected to provide an objective and scalable method for assessing emotional regulation. It will demonstrate how behavioural analytics can be used to evaluate stress management and decision-making in educational settings.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Emotional regulation is the ability of an individual to recognize, monitor, and manage emotional responses in a manner that supports effective decision-making and goal-directed behaviour. It plays a critical role in situations involving uncertainty, time pressure, resource limitations, and unexpected disruptions. Individuals with strong emotional regulation skills are generally capable of maintaining cognitive control, adapting to changing environments, and recovering efficiently from stressful events.

In recent years, behavioural science and cognitive psychology have increasingly focused on digital methods for assessing emotional regulation. Traditional assessment techniques, such as self-report questionnaires and clinical interviews, provide valuable insights but often fail to capture how individuals behave during real-time stressful situations. This has motivated researchers to explore simulation-based environments and serious games as objective assessment platforms capable of measuring behavioral responses continuously.

Simulation-based assessments present users with realistic scenarios where multiple decisions must be taken under varying levels of stress. Rather than relying solely on subjective responses, these systems collect behavioural indicators such as reaction time, decision accuracy, error frequency, persistence after failure, recovery latency, and consistency of choices. These indicators provide measurable evidence of emotional regulation and executive functioning.

The proposed project adopts this simulation-based approach by developing an emergency response game designed to evaluate emotional regulation among higher secondary students (Grades 11–12). The platform introduces controlled stressors such as time constraints, resource shortages, unexpected failures, and conflicting information while continuously recording behavioural data. The collected metrics are analysed to assess four key behavioural

dimensions: Stress Tolerance, Recovery & Resilience, Adaptation & Persistence, and Decision Stability, providing a comprehensive evaluation of emotional regulation.

This literature review examines the psychological foundations of emotional regulation, the cognitive mechanisms associated with stress adaptation, the use of serious games for behavioural assessment, and the existing digital approaches employed for measuring emotional regulation. The findings provide the theoretical basis for designing a scientifically valid behavioural assessment platform.

2.2 Understanding Emotional Regulation

Emotional regulation refers to the processes through which individuals monitor, evaluate, and modify their emotional reactions in order to achieve personal goals and adapt effectively to changing situations. It is a fundamental psychological capability that enables individuals to maintain emotional balance while responding to internal thoughts and external environmental demands. Rather than suppressing emotions, emotional regulation involves managing emotional experiences in a way that supports rational thinking, effective behaviour, and appropriate social interactions.

Psychologists describe emotional regulation as a continuous process that begins with the perception of an emotional event and extends through emotional awareness, cognitive evaluation, behavioural response, and subsequent recovery. Effective regulation allows individuals to remain calm under pressure, recover from unexpected failures, and continue making logical decisions despite experiencing stress or frustration. Conversely, poor emotional regulation may result in impulsive actions, cognitive overload, decreased attention, increased errors, and difficulty adapting to changing circumstances.

One of the most widely accepted theoretical models of emotional regulation was proposed by James Gross through the Process Model of Emotion Regulation. According to this model, emotional regulation can occur before an emotional response is fully generated or after the emotional response has already developed. The model identifies five sequential stages of regulation: situation selection, situation modification, attentional deployment, cognitive reappraisal, and response modulation. These strategies help individuals modify the intensity, duration, or expression of emotions depending on the demands of the situation.

Emotional regulation is closely associated with several higher-order cognitive functions, including attention control, working memory, cognitive flexibility, executive functioning, and

decision-making. During stressful situations, these cognitive processes work together to enable individuals to evaluate available information, prioritize tasks, inhibit impulsive reactions, and adapt their behaviour as circumstances evolve. Consequently, emotional regulation is considered an essential component of effective performance in professions involving emergency management, healthcare, aviation, military operations, and disaster response.

Modern behavioural science recognizes that emotional regulation can be assessed objectively by observing behavioural patterns rather than relying solely on self-reported questionnaires. Digital environments, particularly simulation-based serious games, provide controlled conditions in which behavioural indicators such as reaction time, decision accuracy, panic clicking frequency, recovery latency, error escalation patterns, strategy changes, and decision stability can be continuously recorded. These measurable behavioural features act as indirect indicators of emotional regulation and cognitive resilience.

The proposed emergency response simulation adopts this behavioural assessment approach. Instead of directly asking participants about their emotional state, the system observes how they respond to stressful events such as resource shortages, unexpected emergencies, misinformation, and time constraints. The participant's behavioural responses are evaluated across four behavioural dimensions: Stress Tolerance, Recovery & Resilience, Adaptation & Persistence, and Decision Stability. These dimensions collectively provide an objective estimate of emotional regulation through continuous behavioural observation.

Understanding emotional regulation is therefore essential for designing scientifically valid behavioural assessment systems. By incorporating established psychological principles into interactive simulations, digital platforms can generate objective evidence of how individuals regulate emotions under pressure while providing valuable insights into resilience, stress tolerance, adaptive decision-making, and behavioural recovery.

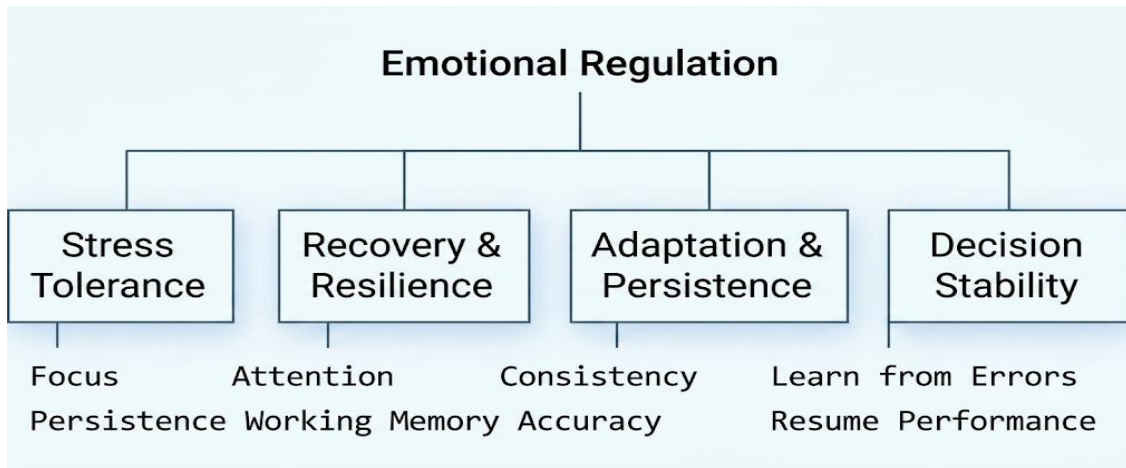


Figure 2.1: Components of Emotional Regulation

Core behavioural components associated with emotional regulation that are measured in simulation-based assessments.

2.3 Psychological Theories of Emotional Regulation

Emotional regulation has been extensively studied in psychology and cognitive science to understand how individuals manage emotional experiences while maintaining effective behaviour. Several theoretical models explain the mechanisms through which emotions influence cognition, behaviour, and decision-making. These theories provide the scientific basis for designing behavioural assessment systems capable of measuring emotional regulation objectively. The proposed emergency response simulation incorporates concepts from these established psychological theories to evaluate users' behavioural responses under controlled stress conditions.

2.3.1 Gross's Process Model of Emotion Regulation

One of the most influential frameworks in emotional regulation research is the Process Model of Emotion Regulation proposed by James Gross. The model explains that emotions develop through a sequence of psychological processes and that regulation can occur at different stages before or after an emotional response is fully expressed.

Gross identified five primary stages of emotional regulation:

1. Situation Selection

Individuals choose situations that are likely to produce desirable emotional outcomes while avoiding situations that may generate negative emotions. This strategy attempts to regulate emotions before they occur.

2. Situation Modification

When stressful situations cannot be avoided, individuals attempt to modify aspects of the environment to reduce emotional impact. For example, requesting additional resources or changing task priorities can reduce stress during problem solving.

3. Attentional Deployment

Individuals selectively focus their attention on information that supports effective decision-making while ignoring distracting or emotionally overwhelming stimuli. Maintaining attention under stress is essential for successful performance in emergency situations.

4. Cognitive Reappraisal

Cognitive reappraisal involves changing the interpretation of a stressful event rather than reacting emotionally to it. Individuals who successfully reframe difficult situations often demonstrate greater resilience and improved decision-making.

5. Response Modulation

This final stage occurs after emotions have already developed. Individuals consciously regulate behavioural responses by controlling facial expressions, impulsive actions, or emotional reactions while continuing task performance.

The proposed simulation reflects this model by presenting users with unexpected challenges such as misinformation, resource shortages, equipment failures, and strict time constraints. Rather than measuring emotional expressions directly, the system evaluates behavioural outcomes including stress tolerance, recovery latency, decision stability, adaptation rate, and persistence after setbacks. These behavioural indicators collectively contribute to the objective assessment of emotional regulation.

2.3.2 Transactional Model of Stress and Coping

The Transactional Model of Stress and Coping, developed by Lazarus and Folkman, explains that stress is not determined solely by external events but by an individual's interpretation of those events. According to this theory, people continuously evaluate environmental demands and compare them with their available coping resources.

The model consists of three major stages:

Primary Appraisal

The individual evaluates whether an event represents a threat, challenge, or neutral situation.

Secondary Appraisal

The individual assesses available resources, skills, and possible actions for dealing with the situation.

Coping Response

Based on these evaluations, behavioural strategies are selected to manage the situation and reduce psychological stress.

In emergency management scenarios, two individuals facing identical situations may display entirely different behavioural responses depending on how they perceive the challenge and evaluate their coping abilities. This explains why behavioural observation provides more reliable evidence than simple questionnaires.

The emergency response simulation applies this theory by introducing progressively difficult situations that require users to continuously reassess priorities, allocate limited resources, and adapt strategies while maintaining performance.

2.3.3 Self-Regulation Theory

Self-Regulation Theory explains how individuals monitor their own behaviour, compare it with desired goals, and adjust actions whenever performance deviates from expectations.

Effective self-regulation requires continuous feedback, planning, monitoring, and behavioural adaptation.

During stressful situations, successful self-regulation enables individuals to:

- Maintain focus despite distractions.
- Correct mistakes after failure.
- Adapt strategies when circumstances change.
- Continue working toward objectives despite obstacles.

Within the proposed simulation, behavioural indicators such as repeated attempts, correction of previous mistakes, strategic adaptation, and persistence serve as observable measures of self-regulation.

2.3.4 Executive Function Theory

Executive functions are higher-order cognitive processes responsible for controlling thoughts, emotions, and behaviour. These functions include working memory, inhibitory control, cognitive flexibility, planning, and decision-making.

Stress often interferes with executive functioning, causing individuals to make impulsive decisions, overlook important information, or abandon tasks prematurely. Conversely, individuals with strong executive control demonstrate greater emotional stability and recover more effectively after setbacks.

The emergency response simulation is designed to challenge executive functions by presenting multiple simultaneous emergencies, incomplete information, changing priorities, and time pressure. Behavioural metrics such as decision accuracy, reaction time, panic clicking frequency, error escalation patterns, and strategy changes provide indirect evidence of executive functioning under stress.

Relevance to the Proposed System

The proposed behavioural assessment platform combines principles from these psychological theories to create scientifically meaningful simulation scenarios. Instead of relying on self-reported emotional experiences, the platform evaluates how users actually behave while facing realistic emergency situations.

Each theory contributes to different aspects of the assessment:

Psychological Theory	Behavioural Dimension Supported	Measured Indicators
Gross's Process Model	Recovery & Resilience	- Recovery latency - Emotional recovery curve - Return to stable performance after setbacks - Persistence after failure
Transactional Model of Stress & Coping	Stress Tolerance	- Decision accuracy under stress - Reaction time under stress - Panic clicking frequency - Error escalation patterns
Self-Regulation Theory	Adaptation & Persistence	- Strategy changes - Retry attempts - Adaptation rate - Task completion despite obstacles
Executive Function Theory	Decision Stability	- Decision consistency - Reaction time - Error escalation patterns - Reduced variability in choices

Together, these theoretical frameworks provide the scientific foundation for measuring emotional regulation through behavioural observation. The simulation therefore serves not merely as a game but as a structured behavioural assessment environment capable of generating objective indicators of emotional regulation, resilience, and adaptive decision-making.

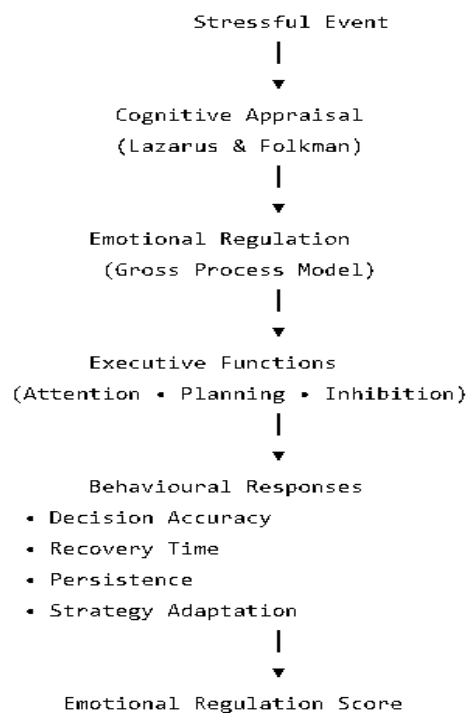


Figure 2.2: Psychological Foundation of the Proposed Assessment

2.4 Emotional Regulation in High-Stress Environments

High-stress environments are situations in which individuals must make accurate decisions while facing uncertainty, limited resources, strict time constraints, and rapidly changing circumstances. Such environments require people to process large amounts of information, prioritize competing tasks, and respond effectively despite experiencing psychological pressure. Emotional regulation becomes particularly important in these situations because emotional reactions directly influence cognitive performance, decision quality, and behavioural stability.

Research in psychology and human factors engineering has shown that stress affects several cognitive functions responsible for effective decision-making. Moderate levels of stress may temporarily enhance alertness and concentration, whereas prolonged or excessive stress often reduces working memory capacity, impairs attention, increases impulsive behaviour, and decreases problem-solving efficiency. Individuals who successfully regulate their emotions are generally better equipped to maintain cognitive performance even under demanding conditions.

Emergency response represents one of the most challenging examples of high-stress decision-making. Emergency personnel, healthcare professionals, firefighters, police officers, military personnel, and disaster management teams frequently operate in environments where information is incomplete, resources are limited, and decisions must be made within seconds. These situations require emotional stability because poor emotional regulation may result in delayed responses, incorrect prioritization, or unsafe decisions that affect both human lives and organizational outcomes.

Similar behavioural demands exist in digital simulation environments designed to reproduce emergency situations. By introducing realistic stressors within a controlled setting, simulation-based systems allow researchers to observe behavioural responses without exposing participants to actual physical danger. This approach has become increasingly popular because it enables standardized assessment while ensuring participant safety and experimental control.

The proposed emergency response simulation adopts this methodology by organizing the assessment into four progressive modules: Baseline Assessment, Stress Induction, Failure & Recovery, and Adaptive Challenge. Each module gradually increases the level of stress while

continuously recording behavioural responses during realistic emergency management scenarios.

Several behavioural indicators can reveal how effectively an individual regulates emotions while interacting with such simulations. These indicators include decision accuracy under stress, reaction time, panic clicking frequency, error escalation patterns, recovery latency, strategy changes, decision stability, adaptation rate, and the emotional recovery curve. Together, these measures provide objective evidence of emotional regulation without requiring participants to explicitly describe their emotional experiences.

One important concept associated with emotional regulation in stressful environments is resilience. Resilience refers to the capacity to recover quickly after setbacks, maintain psychological stability, and continue functioning effectively despite adversity. In behavioural assessment, resilience is reflected through recovery latency, emotional recovery patterns, and improved performance following setbacks, all of which can be measured objectively through simulation event logs.

Another important concept is stress tolerance, which represents an individual's ability to continue performing effectively while experiencing increasing levels of psychological pressure. Stress tolerance can be observed through consistent decision-making, limited performance degradation, reduced panic clicking, and sustained attention during progressively difficult tasks.

The emergency response simulation proposed in this project incorporates both resilience and stress tolerance by gradually increasing environmental complexity throughout the four assessment modules. Participants encounter realistic challenges such as ambulance shortages, equipment failures, communication breakdowns, multiple simultaneous emergencies, and changing priorities. Their behavioural responses are continuously recorded and analysed to estimate emotional regulation using objective behavioural metrics.

Unlike conventional psychological questionnaires that depend on self-reporting, simulation-based assessments observe actual behaviour while individuals interact with stressful situations. This approach improves ecological validity because participant behaviour is measured within realistic task environments rather than hypothetical scenarios. Consequently, the proposed system offers a practical and scientifically grounded method for evaluating emotional regulation among higher secondary students (Grades 11–12).

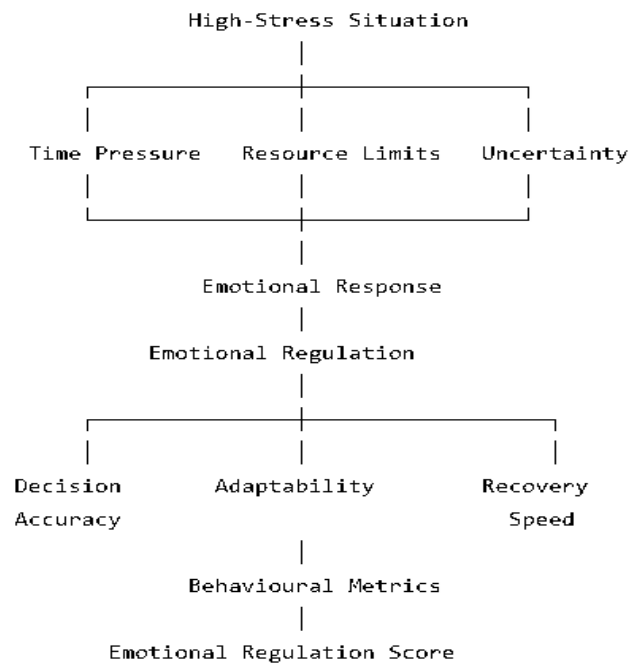


Figure 2.3: Emotional Regulation in High-Stress Environments

Key Behavioural Indicators Measured

Behavioural Metric	What it Indicates
Panic Clicking Frequency	Measures impulsive clicking behaviour under stress
Error Escalation Patterns	Indicates whether errors increase as stress level rises
Strategy Changes	Reflects the ability to modify strategies when situations change
Emotional Recovery Curve	Shows improvement in performance after stressful events
Recovery Latency	Measures the time taken to return to stable performance after a setback
Decision Stability	Indicates consistency of decisions across similar situations
Adaptation Rate	Measures how quickly and effectively one adapts to new challenges

2.5 Serious Games as Behavioural Assessment Tools

Serious games are digital applications designed for purposes beyond entertainment, including education, training, healthcare, and psychological assessment. Unlike conventional video games that primarily focus on recreation, serious games create interactive environments where users perform realistic tasks while the system continuously records behavioural data. These behavioural observations provide objective insights into cognitive abilities, decision-making processes, and emotional responses.

In recent years, serious games have gained significant attention in behavioural science because they enable researchers to assess human performance under controlled yet realistic conditions. Traditional assessment methods, such as questionnaires and interviews, rely heavily on self-reported responses, which may be influenced by memory bias, social desirability, or inaccurate self-perception. Serious games overcome these limitations by measuring actual behaviour during task execution, making assessments more objective and ecologically valid.

One of the major advantages of serious games is their ability to simulate complex real-world environments without exposing participants to physical risk. Emergency response, disaster management, healthcare, military operations, and aviation training commonly use simulation-based systems to evaluate decision-making, adaptability, and performance under pressure. These simulations recreate realistic situations where users must respond to changing conditions while balancing multiple priorities.

Modern serious games collect a wide range of behavioural metrics during gameplay. These include reaction time, decision accuracy, panic clicking frequency, error escalation patterns, strategy changes, recovery latency, emotional recovery curve, decision stability, and adaptation rate. Advanced systems analyse these behavioural indicators to estimate psychological constructs such as stress tolerance, recovery & resilience, adaptation & persistence, decision stability, executive functioning, and emotional regulation.

Behavioural assessment through serious games offers several advantages:

- Continuous observation without interrupting the participant.
- Objective measurement of real-time behaviour.
- Controlled introduction of stressful events.

- Repeatable scenarios for consistent evaluation.
- Automatic collection and analysis of behavioural data.

The proposed emergency response simulation follows the principles of serious game design by creating realistic emergency scenarios that require participants to allocate resources, prioritize incidents, and respond to unexpected disruptions. The assessment is organized into four progressive modules—Baseline Assessment, Stress Induction, Failure & Recovery, and Adaptive Challenge—to observe behavioural changes under increasing levels of stress. Rather than measuring gaming performance alone, the platform evaluates behavioural patterns that reflect emotional regulation. Events such as ambulance shortages, misinformation, equipment failures, and multiple simultaneous emergencies are intentionally introduced to observe how participants adapt their strategies, recover from setbacks, and maintain stable decision-making.

The behavioural data collected throughout gameplay forms the basis of the emotional regulation assessment. Metrics such as recovery latency, emotional recovery curve, panic clicking frequency, error escalation patterns, strategy changes, decision stability, and adaptation rate provide objective evidence of emotional regulation. These behavioural indicators are combined using a weighted scoring framework to estimate Stress Tolerance, Recovery & Resilience, Adaptation & Persistence, and Decision Stability. By integrating serious game mechanics with behavioural analytics, the proposed system becomes both an engaging simulation and a scientifically grounded assessment platform.

Table 2.2 Advantages of Serious Games for Behavioural Assessment

<i>Traditional Assessment</i>	<i>Serious Game Assessment</i>
<i>Self-reported responses</i>	Behaviour observed through interaction
<i>Static questionnaire</i>	Interactive simulation scenarios
<i>Single assessment point</i>	Continuous monitoring
<i>Subjective evaluation</i>	Objective behavioural metrics
<i>Limited behavioural evidence</i>	Event logging and analytics
<i>No real-time adaptation</i>	Progressive stress induction through multiple modules
<i>Manual scoring</i>	Automated scoring and behavioural analytics

2.6 Existing Emotional Regulation Assessment Methods and Research Gap

Emotional regulation has traditionally been assessed using psychological questionnaires, clinical interviews, laboratory-based experiments, and physiological measurements. Each of these approaches provides valuable information regarding emotional functioning; however, they also possess limitations that reduce their effectiveness in evaluating real-world behaviour under stressful conditions.

One of the most commonly used assessment techniques involves self-report questionnaires. Instruments such as the Emotion Regulation Questionnaire (ERQ) and the Difficulties in Emotion Regulation Scale (DERS) require individuals to evaluate their own emotional experiences and coping strategies. Although these questionnaires are easy to administer and widely validated, they depend heavily on self-awareness and honesty. Participants may unintentionally provide inaccurate responses due to memory bias, misunderstanding of questions, or the desire to present themselves positively.

Clinical interviews conducted by psychologists provide more detailed assessments by exploring emotional experiences through structured conversations. While interviews allow professionals to identify behavioural patterns and emotional difficulties, they require trained clinicians, consume considerable time, and are often difficult to standardize across large populations. As a result, interviews are less suitable for large-scale behavioural screening.

Researchers have also explored physiological measurements to assess emotional regulation. Techniques such as electroencephalography (EEG), electrocardiography (ECG), heart rate variability (HRV), galvanic skin response (GSR), and eye-tracking measure biological responses associated with stress and emotional arousal. Although these methods provide objective physiological data, they require specialized equipment, controlled laboratory environments, and trained personnel. Such requirements increase implementation costs and reduce accessibility for educational institutions and routine behavioural assessments.

In recent years, simulation-based behavioural assessment has emerged as an alternative approach. Rather than directly measuring emotions, these systems evaluate observable behaviours while individuals perform realistic tasks. Serious games and interactive simulations continuously record behavioural metrics such as reaction time, decision accuracy, panic clicking frequency, error escalation patterns, recovery latency, strategy changes, decision stability, adaptation rate, and the emotional recovery curve. These behavioural indicators provide indirect yet objective evidence of emotional regulation in environments that closely resemble real-life situations.

Despite these advances, many existing simulation systems primarily focus on evaluating technical skills, task performance, or procedural knowledge. Relatively few platforms are specifically designed to assess emotional regulation through behavioural analytics. Furthermore, many available systems depend on expensive hardware, physiological sensors, or highly specialized laboratory settings, limiting their practical use in educational environments. The proposed emergency response simulation addresses these limitations by providing a software-based behavioural assessment platform that requires only a standard computer. The assessment is organised into four progressive modules—Baseline Assessment, Stress Induction, Failure & Recovery, and Adaptive Challenge—to observe behavioural responses under increasing levels of stress. Instead of relying on physiological sensors or subjective questionnaires, the system evaluates emotional regulation by analysing behavioural responses to controlled stressors embedded within realistic emergency scenarios. Behavioural indicators such as panic clicking frequency, error escalation patterns, recovery latency, strategy changes, decision stability, adaptation rate, and emotional recovery curve are automatically recorded and analysed using a weighted scoring framework.

This approach offers several advantages. It enables objective behavioural assessment, supports repeated evaluations under standardized conditions, reduces dependence on costly

hardware, and provides immediate analytical feedback. Consequently, the proposed system represents a practical, scalable, and scientifically grounded method for assessing emotional regulation among undergraduate students.

Table 2.3 Comparison of Existing Assessment Methods

<i>Assessment Method</i>	Advantages	Limitations
<i>Self-report Questionnaires</i>	Easy to administer, inexpensive	Subjective, affected by response bias
<i>Clinical Interviews</i>	Detailed psychological evaluation	Time-consuming, requires experts
<i>Physiological Sensors (EEG, HRV, GSR)</i>	Objective physiological data	Expensive, requires specialized equipment
<i>Simulation-Based Assessment</i>	Continuous behavioural observation in realistic environments	Often measures performance rather than emotional regulation
<i>Proposed System</i>	Software-based, objective behavioural analytics, four-module assessment, automated scoring	Requires carefully designed simulation scenarios

Research Gap

Current emotional regulation assessment techniques either depend on subjective self-reporting or require expensive physiological equipment. Existing simulation platforms generally emphasise task completion, processing speed, or technical performance rather than behavioural dimensions such as Stress Tolerance, Recovery & Resilience, Adaptation & Persistence, and Decision Stability. There is limited research on software-based systems that objectively measure these behavioural constructs using structured event logging and behavioural analytics. There remains a need for a cost-effective, software-based assessment platform capable of objectively measuring emotional regulation through behavioural analytics in realistic, stress-inducing environments.

The proposed emergency response simulation addresses this gap by integrating controlled stressors, structured behavioural event logging, a four-module assessment framework, and a weighted behavioural scoring model within a serious game environment. This approach enables continuous observation of behavioural responses while maintaining accessibility, scalability, and scientific validity.

2.7 Summary

This literature review examined the theoretical and technological foundations relevant to the proposed behavioural assessment platform. It discussed the concept of emotional regulation, its significance in maintaining cognitive control during stressful situations, and the major psychological theories that explain how individuals regulate emotions and adapt to changing environments. The review also highlighted the importance of emotional regulation in high-stress domains such as emergency response, where effective decision-making and behavioural stability are essential.

Furthermore, the literature explored the growing role of serious games and simulation-based environments in behavioural assessment. Unlike traditional self-report questionnaires or laboratory-based physiological measurements, serious games provide realistic and controlled settings that enable continuous observation of user behaviour. Behavioural indicators such as decision accuracy, persistence, recovery time, and adaptability can be objectively measured, making simulation-based assessment a promising approach for evaluating emotional regulation.

The review of existing assessment methods identified several limitations, including dependence on subjective responses, the need for specialized equipment, and limited applicability in real-world educational settings. Although recent simulation platforms have improved behavioural assessment, many primarily evaluate task performance rather than emotional regulation.

Based on these findings, the proposed emergency response simulation addresses an important research gap by providing a software-based behavioural assessment platform that objectively evaluates emotional regulation through realistic emergency scenarios and behavioural analytics. The system incorporates a four-module assessment framework, structured SDK-compatible event logging, and a weighted scoring model based on Stress Tolerance, Recovery

& Resilience, Adaptation & Persistence, and Decision Stability. This approach offers a scientifically grounded, scalable, and accessible method for assessing emotional regulation among higher secondary students (Grades 11–12). The insights gained from this literature review provide the theoretical foundation for the design, implementation, and evaluation of the proposed system presented in the subsequent chapters.

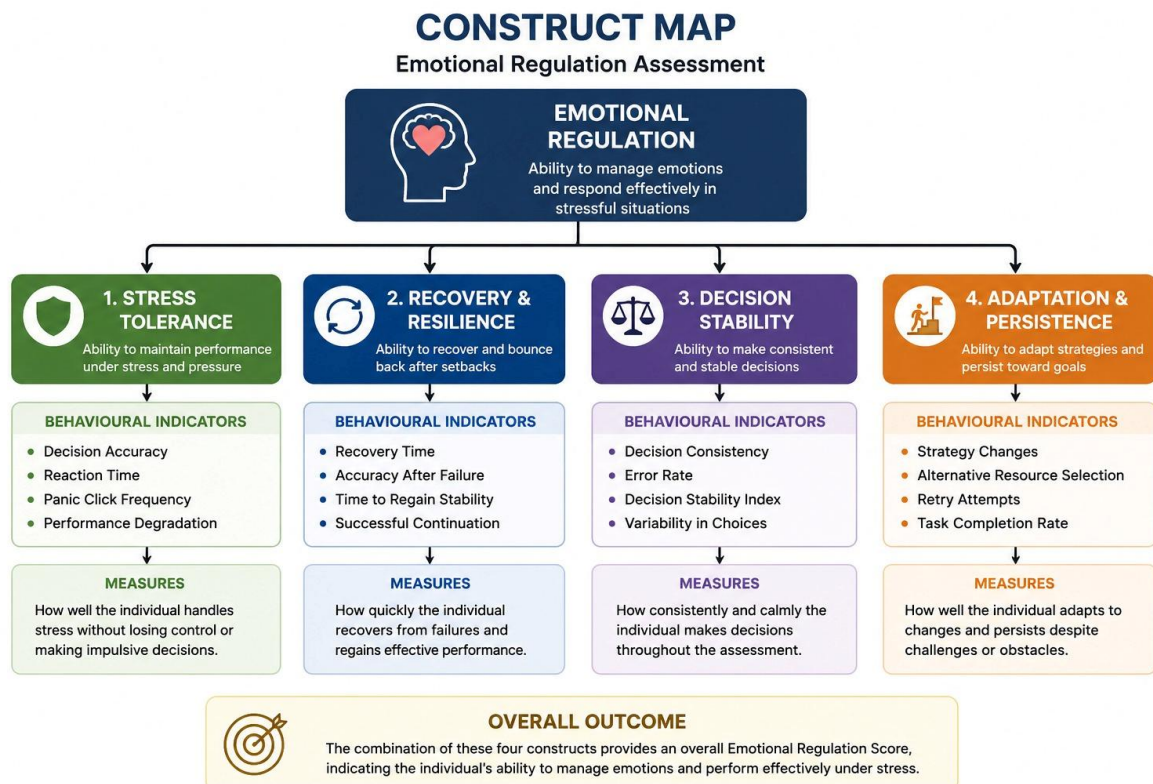
CHAPTER 3

CONSTRUCT MAP

3.1 Introduction

The proposed system evaluates *Emotional Regulation* through observable behavioural responses during emergency response simulations. Instead of relying on questionnaires, the system measures how participants react, recover, and adapt when exposed to controlled stressful situations. The construct map identifies the key psychological dimensions assessed and their corresponding behavioural indicators.

3.2 Construct Map



3.3 Behavioural Dimensions

The emotional regulation construct is divided into four behavioural dimensions.

1. Stress Tolerance

Measures the participant's ability to maintain performance under stressful conditions.

Indicators:

- Decision accuracy
- Reaction time
- Panic click frequency

2. Recovery & Resilience

Measures how quickly the participant recovers after failures or unexpected challenges.

Indicators:

- Recovery time
- Accuracy after failure
- Successful task continuation

3. Decision Stability

Measures the consistency of decision-making throughout the assessment.

Indicators:

- Decision consistency
- Error frequency
- Stability under pressure

4. Adaptation & Persistence

Measures the participant's ability to adjust strategies and continue despite setbacks.

Indicators:

- Strategy changes
- Retry attempts
- Task completion

3.4 Construct Mapping Table

CONSTRUCT MAPPING TABLE

Emotional Regulation Assessment Framework

Construct	Definition	Behavioural Indicators	Mapped Events (Examples)	Measurement Purpose
 1. STRESS TOLERANCE	Ability to maintain performance and make appropriate decisions under stress and pressure.	- Decision Accuracy - Reaction Time - Panic Click Frequency - Performance Degradation	- click - wrong_decision - timeout - stress_trigger - interrupt	Measures how well the individual handles stress without losing control or making impulsive decisions.
 2. RECOVERY & RESILIENCE	Ability to recover quickly from setbacks, failures or unexpected challenges and regain stable performance.	- Recovery Time - Accuracy After Failure - Time to Regain Stability - Successful Continuation	- wrong_decision - resource_shortage - module_restart - correct_decision - task_complete	Measures how quickly the individual bounces back after failures and continues effectively.
 3. DECISION STABILITY	Ability to make consistent, rational and stable decisions throughout the assessment.	- Decision Consistency - Error Rate - Decision Stability Index - Variability in Choices	- click - wrong_decision - correct_decision - resource_change	Measures consistency and stability in decision-making across different situations and stress levels.
 4. ADAPTATION & PERSISTENCE	Ability to adapt strategies, utilize alternative resources and persist in completing tasks despite obstacles.	- Strategy Changes - Resource Adaptation - Retry Attempts - Task Completion Rate	- resource_change - strategy_change - retry - task_complete - module_complete	Measures adaptability and persistence in continuing and completing tasks despite challenges.



Overall Outcome: These four constructs collectively determine the Emotional Regulation Score by analysing behavioural responses under varying stress conditions in the simulation.

3.5 Summary

The construct map provides the foundation for the behavioural assessment framework. Each psychological construct is measured using objective behavioural indicators collected during the emergency response simulation. These measurements are later analysed to generate the overall Emotional Regulation Score. The developed application implements all behavioural constructs identified in the construct map through gameplay interactions and event logging.

CHAPTER 4

SYSTEM FLOW DIAGRAM

4.1 Introduction

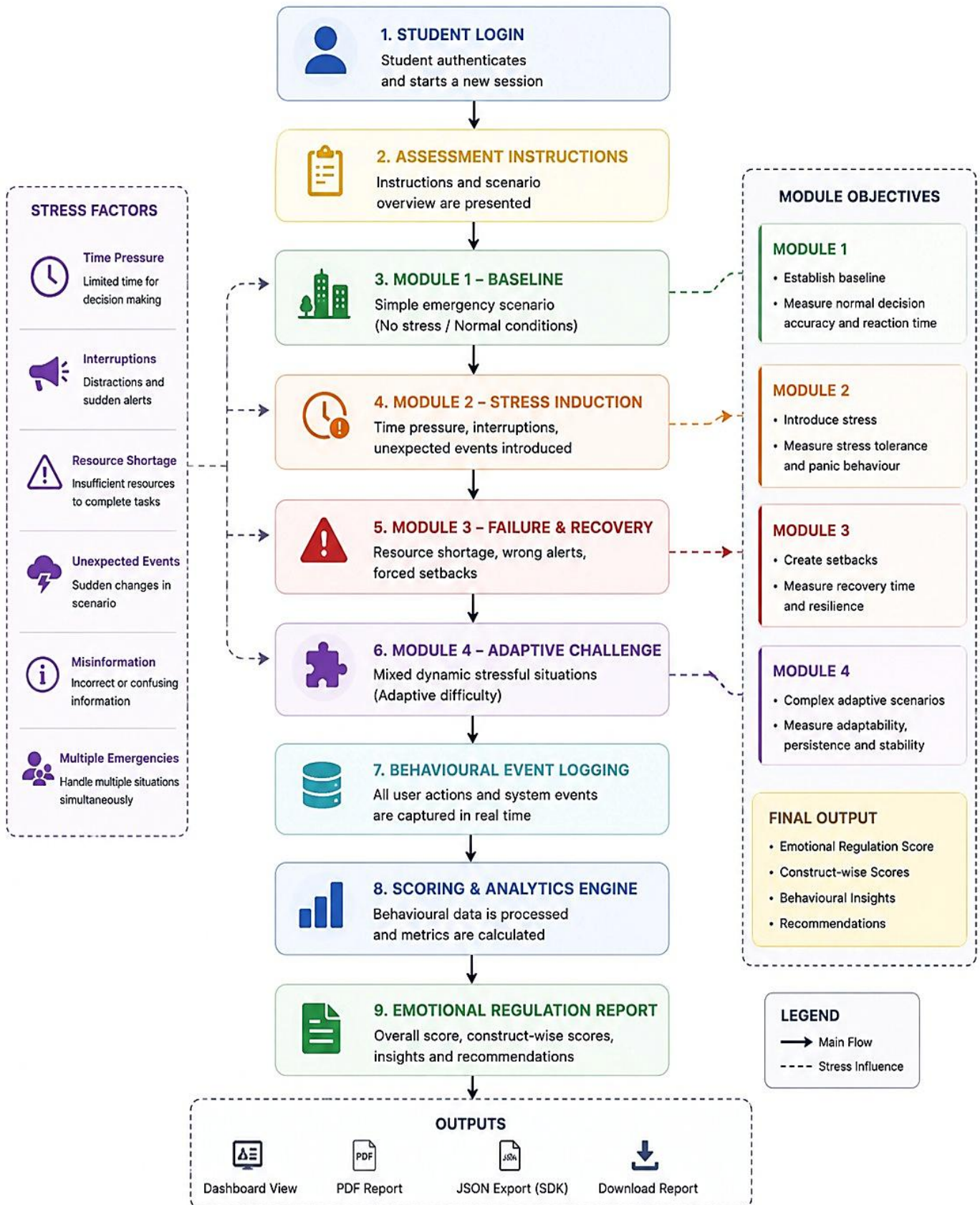
The proposed system follows a structured workflow to evaluate emotional regulation through an emergency response simulation. The assessment is divided into four modules, each designed to observe different behavioural responses under varying stress levels. Throughout the assessment, user interactions are continuously recorded and analysed to generate an Emotional Regulation Score.

4.2 System Workflow

The overall workflow of the proposed system is illustrated below.

SYSTEM WORKFLOW

Emotional Regulation Assessment Platform



4.3 Workflow Description

The assessment begins with student authentication and presentation of assessment instructions. The participant then completes four sequential modules:

Module 1 – Baseline: Measures normal decision-making without external stress.

Module 2 – Stress Induction: Introduces controlled stressors such as reduced time and interruptions.

Module 3 – Failure & Recovery: Forces unavoidable setbacks to evaluate recovery and resilience.

Module 4 – Adaptive Challenge: Combines multiple stress events to assess persistence and decision stability.

During all modules, the system records behavioural events including reaction time, decision accuracy, errors, and strategy changes. These events are processed by the scoring engine to generate the final assessment report.

4.4 Assessment Flow

ASSESSMENT FLOW

Step	Activity	Description	Input / Trigger	Output	Logged Events
1 	Student Login	Student authenticates using credentials and starts a new session.	Student ID, Password	Session Created	- login_start - session_start
2 	Instructions	Assessment instructions and scenario overview are presented.	Session ID	Assessment Started	- instructions_viewed - assessment_start
3 	Module 1 – Baseline	Simple emergency scenario in normal conditions to establish baseline behaviour.	Module 1 Start	Baseline Performance Data	- module_start (1) - decision / action events - module_complete (1)
4 	Module 2 – Stress Induction	Time pressure, distractions and unexpected events are introduced.	Module 2 Start	Stress Behaviour Data	- stress_trigger - decision / action events - module_complete (2)
5 	Module 3 – Failure & Recovery	Resource shortages and forced setbacks to evaluate recovery and resilience.	Module 3 Start	Recovery Behaviour Data	- failure_event - recovery_actions - module_complete (3)
6 	Module 4 – Adaptive Challenge	Mixed and dynamic stressful situations to assess adaptation, persistence and stability.	Module 4 Start	Adaptive Behaviour Data	- stress_trigger - strategy_change - module_complete (4)
7 	Event Analysis	All logged events are processed to extract behavioural metrics.	All Module Event Logs	Behavioural Metrics	metrics_generated
8 	Score Generation	Metrics are mapped to constructs and weighted to generate overall Emotional Regulation Score.	Behavioural Metrics	Construct Scores & Overall Score	score_computed
9 	Report & Feedback	Comprehensive assessment report with insights and recommendations is generated.	Overall Score & Construct Scores	Assessment Report (Dashboard / PDF / JSON)	- report_generated - session_end

4.5 Summary

The system workflow ensures that each participant experiences a structured assessment process, progressing from normal conditions to increasingly stressful situations. Continuous behavioural logging enables objective measurement of emotional regulation and supports the generation of a comprehensive assessment report.

CHAPTER 5

EVENT LOGGING & SDK SCHEMA

5.1 Introduction

To objectively assess emotional regulation, the system records every user interaction during the emergency response simulation. These interactions are stored as structured event logs and later analyzed to measure behavioural indicators such as stress tolerance, recovery, decision stability, and persistence. The collected data also follows an SDK-compatible format, allowing future integration with analytics or AI models.

5.2 Event Logging

Every meaningful action performed by the participant is recorded as an event. These events include user actions, system-generated stress triggers, and task outcomes.

Examples of logged events include:

- * User login
- * Module started
- * Resource allocated
- * Wrong decision
- * Timeout
- * Stress trigger activated
- * Strategy changed
- * Module completed

Each event contains a timestamp and session information to enable accurate behavioural analysis.

5.3 Event Schema

The following table describes the structure of each logged event.

5.3 Event Schema

The following table describes the structure of each logged event captured during the assessment.

Field	Type	Description	Example	Mandatory
 eventId	String	Unique identifier for the event.	"EVT10245"	✓
 studentId	String	Unique identifier of the student.	"STU001"	✓
 sessionId	String	Unique identifier for the assessment session.	"SES001"	✓
 module	Integer	Current module number (1 to 4).	2	✓
 scenarioId	String	Identifier of the current scenario.	"SCN02"	✓
 eventType	String	Type of event captured. (Click, Timeout, Stress Trigger, Module Start, etc.)	"stress_trigger"	✓
 action	String	Action performed by the user.	"allocate_resource"	✓
 response	String	User selected option or resource.	"Medical_Team"	✓
 correct	Boolean	Indicates whether the action/response was correct. (true = correct, false = incorrect)	false	✓
 reactionTimeMs	Integer	Time taken by the user to respond (in milliseconds).	2450	(Conditional)
 stressLevel	String	Stress level at the time of event. (Low / Medium / High)	"High"	✓
 timestamp	DateTime	Date and time when the event occurred (UTC).	"2025-05-20T10:15:30.450Z"	✓
 metadata	Object	Additional contextual information related to the event (e.g., resource state, difficulty, remaining time).	{ "remainingTime": 120, "resourcesLeft": 3 }	(Optional)

5.4 Behavioural Metrics

The recorded events are used to calculate the following behavioural metrics.

5.4 Behavioural Metrics

The recorded events are used to calculate the following behavioural metrics.

Metric	Description	How It Is Calculated	Unit	Purpose
 1. Decision Accuracy	Measures the percentage of correct decisions made by the participant.	$\frac{\text{Number of Correct Decisions}}{\text{Total Decisions}} \times 100$	Percentage (%)	Indicates overall effectiveness in decision making.
 2. Reaction Time	Measures the average time taken to respond to a prompt or situation.	$\frac{\text{Sum of All Reaction Times (ms)}}{\text{Total Number of Responses}}$	Milliseconds (ms)	Indicates speed of information processing and response under pressure.
 3. Panic Click Frequency	Counts the number of rapid incorrect clicks or random actions under stress.	Total Incorrect/Random Quick Clicks during High Stress Events	Count (Frequency)	Indicates impulsivity and inability to stay calm under stress.
 4. Recovery Time	Measures the time taken to return to stable performance after a failure or setback.	Time from Failure Event until Next Correct Decision	Seconds (s)	Indicates ability to recover quickly from setbacks.
 5. Strategy Changes	Counts the number of times the participant changes approach or resource to solve a problem.	Total Number of Distinct Strategy/Resource Changes	Count (Frequency)	Indicates adaptability and flexibility in changing approaches.
 6. Error Escalation	Measures the increase in errors as stress level increases.	$\frac{\text{Errors in High Stress} - \text{Errors in Low Stress}}{\text{Errors in Low Stress}}$	Ratio (Index)	Indicates how performance degrades under increasing stress.
 7. Persistence Score	Measures the ability to continue and complete tasks despite challenges and setbacks.	$\frac{(\text{Tasks Completed Successfully Despite Setbacks})}{\text{Total Tasks Attempted}} \times 100$	Percentage (%)	Indicates determination and ability to persevere until completion.

5.5 Scoring Model

The final Emotional Regulation Score is calculated using four behavioural dimensions.

Dimension	Weight
Recovery & Resilience	30%
Stress Tolerance	25%
Adaptation & Persistence	25%
Decision Stability	20%

The weighted scores are combined to generate the overall Emotional Regulation Score.

5.6 SDK-Compatible JSON Format

The assessment results are exported in a structured JSON format for future integration.

json

```
{
  "studentId": "STU001",
  "sessionId": "SES001",
  "overallScore": 84,
  "subScores": {
    "stressTolerance": 80,
    "recoveryResilience": 88,
    "adaptationPersistence": 82,
    "decisionStability": 86
  },
  "behaviouralMetrics": {
    "reactionTime": 2400,
    "panicClicks": 2,
    "recoveryTime": 8,
    "strategyChanges": 3
  }
}
```

5.7 Summary

The event logging framework forms the core of the proposed assessment system. By recording user interactions in a structured format, the platform can objectively measure behavioural responses under stress and generate reliable emotional regulation scores. The SDK-compatible schema also supports future integration with AI-based analytics and external assessment platforms.

CHAPTER 6

MVP SYSTEM ARCHITECTURE

6.1 Introduction

This chapter presents the Minimum Viable Product (MVP) architecture of the proposed **Crisis Dispatcher** system. The application is developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)** and follows a modular architecture to support simulation-based behavioural assessment. The system separates the presentation layer, application logic, and data management, enabling efficient event logging, behavioural analytics, psychological scoring, and administrative monitoring.

The architecture is designed to provide a scalable and maintainable platform where students participate in a serious game while the system continuously records behavioural interactions. The collected behavioural data are analysed using a scoring engine to evaluate emotional regulation across four assessment modules.

6.2 System Overview

The proposed system follows a three-tier architecture consisting of the Frontend, Backend, and Database layers. Each layer performs specific responsibilities while communicating through secure REST APIs.

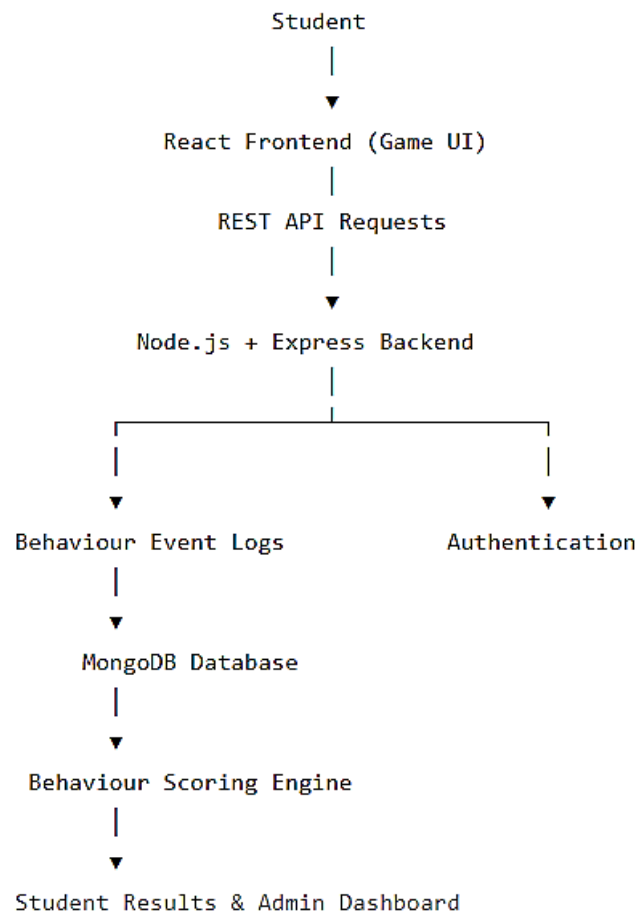
The architecture consists of:

- **Frontend (React.js):** Provides the interactive serious game interface where students complete the emergency dispatch simulation.
- **Backend (Node.js & Express.js):** Handles authentication, event logging, behavioural analysis, and score generation.
- **Database (MongoDB):** Stores user information, behavioural event logs, assessment sessions, and final assessment scores.

The overall workflow begins when a student starts an assessment. During gameplay, every interaction is recorded as behavioural events and stored in MongoDB. Once the assessment is

completed, the scoring engine processes the recorded events and generates behavioural scores that are displayed to both students and administrators.

Figure 6.1 MVP System Architecture



6.3 Frontend Architecture

The frontend is implemented using **React.js** and Vite. It is responsible for rendering the interactive emergency dispatch simulation and capturing user interactions in real time.

The frontend architecture consists of the following modules:

Authentication Module

Provides login and registration functionality for students and administrators using JWT authentication.

Game Engine Module

The Game Engine controls the complete simulation and consists of several reusable components.

- **Command Centre Component**
 - Controls the assessment flow.
 - Generates emergency scenarios.

- Manages timers.
- Controls module transitions.
- **Radar Map Component**
 - Displays active emergency incidents.
 - Shows incident locations.
- **Incident Feed Component**
 - Displays incoming emergency cards.
 - Shows incident descriptions and priorities.
- **Resource Bay Component**
 - Displays available emergency resources.
 - Allows players to dispatch units.
- **HUD (Heads-Up Display)**
 - Displays:
 - Current incidents
 - Resources remaining
 - Lives saved
 - Lives lost
 - Time remaining
 - Mission progress

Student Module

Provides:

- Assessment page
- Results page
- Behavioural score visualization

Administrator Module

Provides:

- Student management
- Assessment reports
- Behavioural analytics
- Event log viewer

6.4 Backend Architecture

The backend is developed using **Node.js** and **Express.js** and follows the MVC (Model-View-Controller) architecture.

The backend is responsible for:

- User authentication
- Behaviour event collection
- Session management
- Behavioural score calculation
- Database communication
- Administrative analytics

The backend contains the following components.

Controllers

Authentication Controller

Handles:

- Registration
- Login
- JWT generation
- User verification

Assessment Controller

Handles:

- Assessment initialization
- Event submission
- Session completion
- Score generation

Middleware

Authentication middleware validates JWT tokens and protects secured API endpoints.

Routes

REST APIs are organized into separate modules.

Examples include:

- Authentication API
- Assessment API
- Event API
- Score API

Services

The behavioural scoring engine is implemented as a separate service layer.

Its responsibilities include:

- Reading behavioural events
- Calculating behavioural metrics
- Normalizing raw values
- Applying weighted scoring
- Generating the final emotional regulation score

Configuration

All behavioural scoring thresholds are stored inside a configuration file.

Examples include:

- Minimum reaction time
- Maximum reaction time
- Panic click threshold
- Recovery latency threshold
- Decision stability threshold
- Adaptation threshold

Keeping these values configurable allows future modifications without changing the scoring algorithm.

6.5 Database Design

MongoDB is used to store all application data.

The database consists of three primary collections.

User Collection

Stores user account information.

Main attributes include:

- Name
- Email
- Password (hashed)
- User Role (Student/Admin)

Event Collection

Stores every behavioural interaction generated during gameplay.

Each event contains:

- Session ID
- Student ID
- Module Number
- Scenario ID
- Event Type
- Timestamp
- Reaction Time
- Decision
- Correct Decision
- Panic Click Count
- Retry Count
- Strategy Change
- Recovery Latency
- Decision Stability
- Adaptation Rate
- Metadata

The Event Collection forms the behavioural dataset used by the scoring engine.

Score Collection

Stores the final assessment results.

Each record contains:

- Overall Score
- Recovery & Resilience Score
- Stress Tolerance Score
- Adaptation & Persistence Score
- Decision Stability Score
- Behavioural Metrics
- Gameplay Statistics

6.6 Assessment Workflow

The behavioural assessment follows a structured workflow consisting of four stages.

Step 1 – Assessment Initialization

The student logs into the system and starts the assessment.

A unique **Session ID** is generated for the assessment.

Step 2 – Real-Time Behaviour Logging

During gameplay, the Command Centre continuously records behavioural interactions including:

- Resource selection
- Dispatch decisions
- Reaction time
- Decision changes
- Panic clicks
- Retry attempts
- Timeout events
- Dynamic adaptations

Every event is immediately stored in MongoDB.

Step 3 – Behavioural Score Calculation

After completing Module 4, the backend retrieves all behavioural events associated with the Session ID.

The Behavioural Scoring Engine then:

- Calculates behavioural metrics
- Normalizes metric values
- Applies weighted scoring
- Generates the final emotional regulation score

The scoring model uses the following weights:

<i>Behavioural Dimension</i>	<i>Weight</i>
<i>Recovery & Resilience</i>	30%
<i>Stress Tolerance</i>	25%
<i>Adaptation & Persistence</i>	25%
<i>Decision Stability</i>	20%

Step 4 – Results and Analytics

The generated scores are stored in the database.

Students can view:

- Overall Emotional Regulation Score
- Module-wise performance
- Behavioural metrics

Administrators can access:

- Student assessment reports
- Behavioural analytics
- Event logs
- Overall assessment statistics

6.7 Advantages of the Proposed MVP Architecture

The proposed architecture provides several advantages for behavioural assessment and future system development.

The major advantages include:

- Modular and scalable architecture.
- Clear separation between frontend, backend, and database layers.
- Real-time behavioural event logging.
- Secure JWT-based authentication.
- Configurable behavioural scoring engine.
- Efficient MongoDB data storage.
- Support for behavioural analytics.
- Easy integration of future assessment modules.
- Maintainable and reusable software components.
- Suitable for educational and research-based behavioural assessment.

6.8 Summary

This chapter presented the MVP architecture of the proposed **Crisis Dispatcher** system. The architecture adopts the MERN stack to provide a scalable and modular platform for serious game-based behavioural assessment. The frontend manages the emergency dispatch simulation, while the backend performs authentication, behavioural event processing, and psychological score calculation. MongoDB stores behavioural logs and assessment results for future analysis. This architecture provides a strong foundation for implementing an interactive behavioural assessment platform capable of evaluating emotional regulation through realistic emergency response simulations.

CHAPTER 7

PROTOTYPE SCREENS

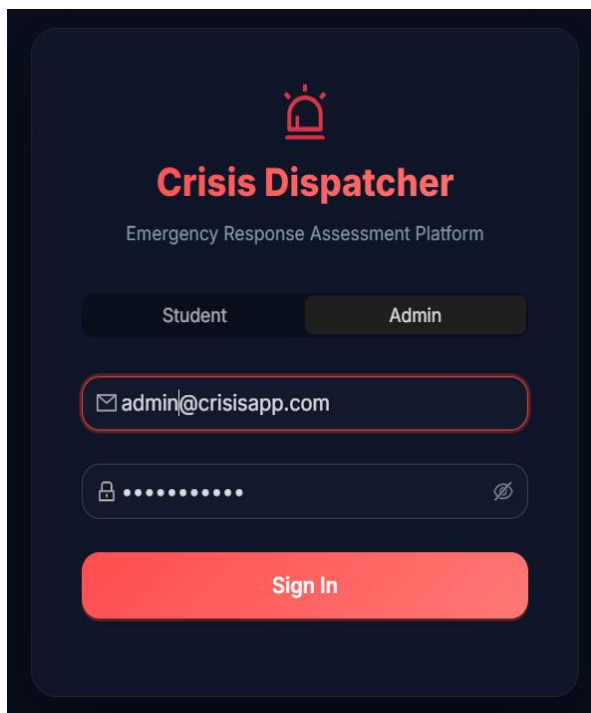


Fig 7.1

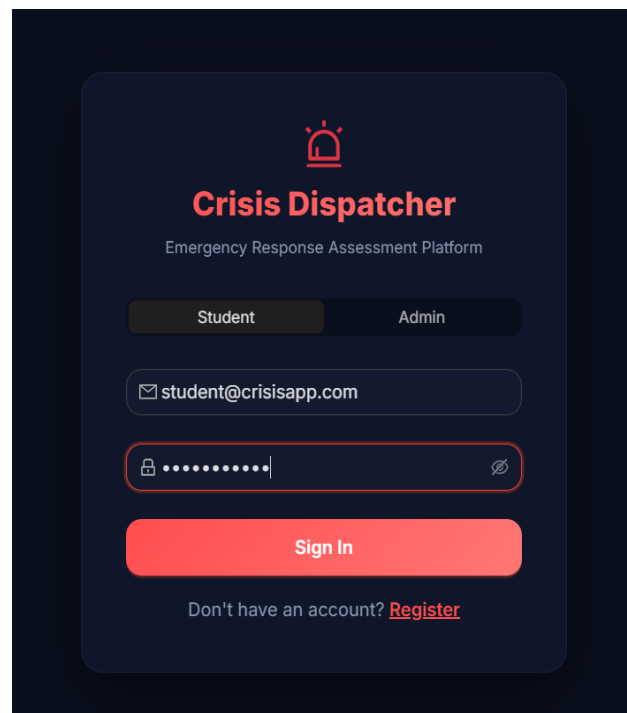


Fig 7.2

Figure 7.1 Login Page (Admin)

The Admin Login page provides secure authentication for administrators using email and password credentials. After successful login, administrators can access the dashboard to monitor student assessments, behavioural analytics, and event logs.

Figure 7.2 Login Page (Student)

The Student Login page allows registered students to securely access the Crisis Dispatcher assessment platform. Successful authentication redirects students to their personalized dashboard.

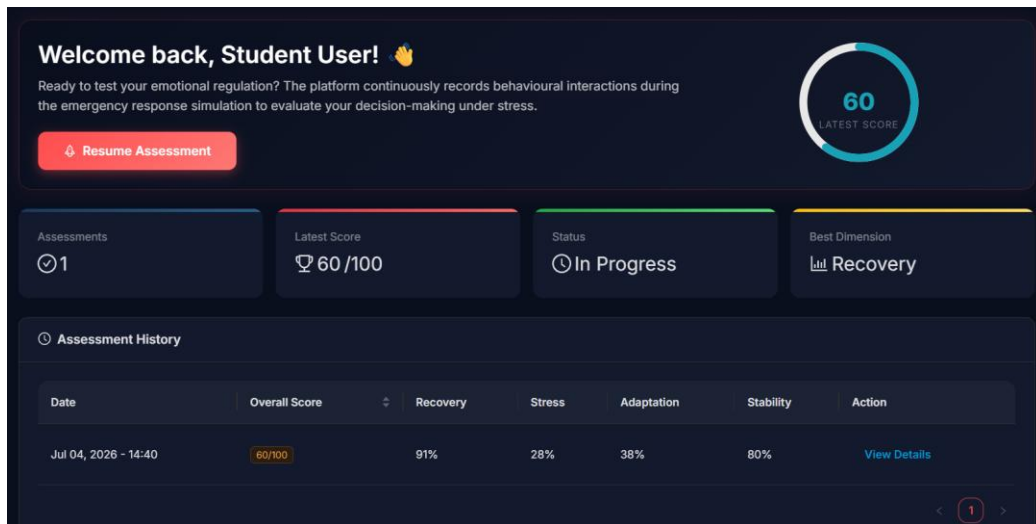


Figure 7.3 Student Dashboard

The Student Dashboard serves as the main interface where students can view assessment status, access instructions, start the assessment, and review their previous results. It provides an organized entry point to the behavioural assessment process.

The Crisis Dispatcher registration page has a dark blue background. At the top, a red bell icon is followed by the title "Crisis Dispatcher" and the subtitle "Create your student account". The form includes four input fields: "Full name", "Email address", "Password (min 6 characters)", and "Confirm password". Red error messages are displayed below the email and password fields: "Please enter your email" and "Please enter a password". A red "Create Account" button is located below the form fields. At the bottom, a link reads "Already have an account? [Sign In](#)".

Figure 7.4 Registration Page

The Registration page enables new students to create an account by entering their personal information and login credentials. The system validates the submitted information before creating the user account.

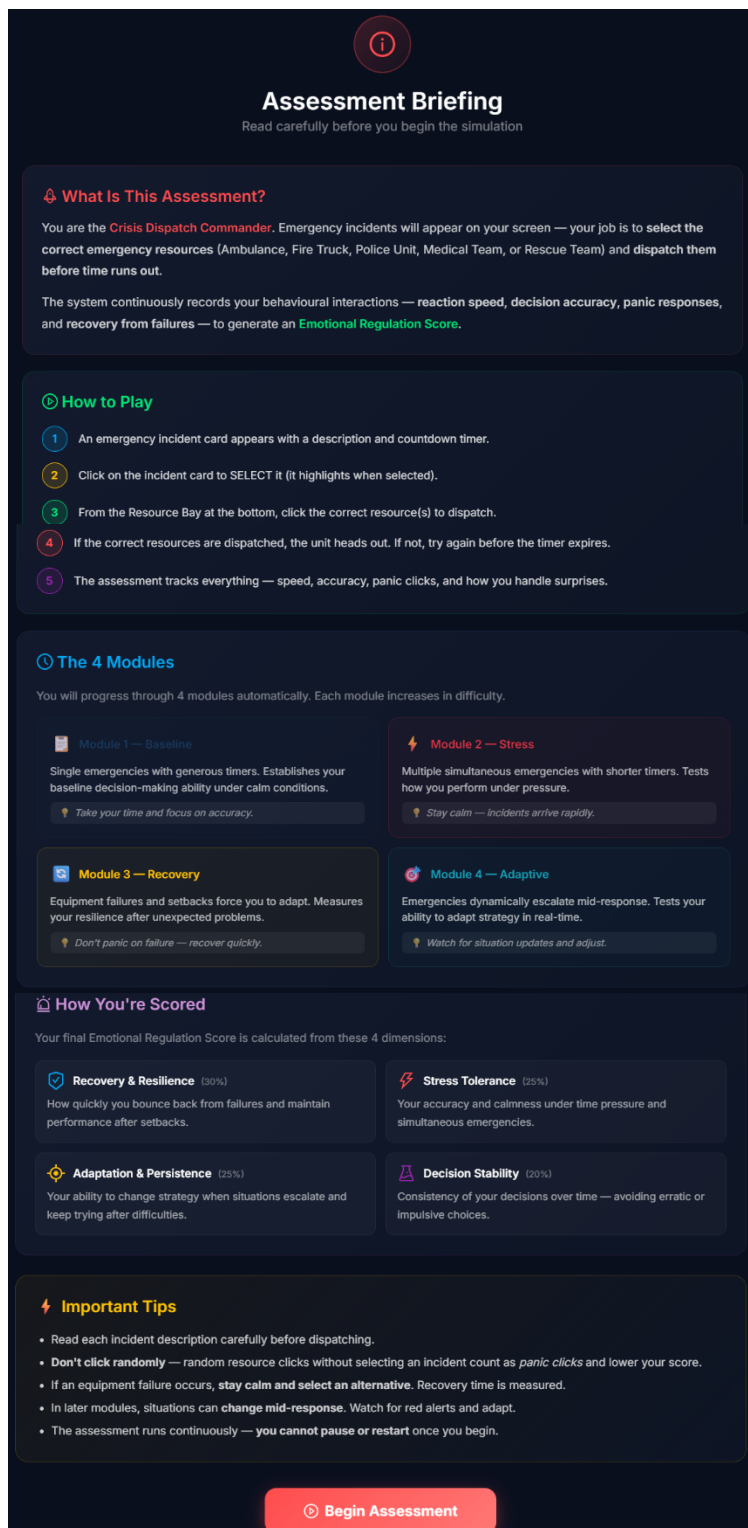


Figure 7.5 Instructions / Guidelines Page

The Instructions page presents the assessment objectives, gameplay rules, scoring guidelines, and general instructions to students before starting the simulation. This ensures that all participants clearly understand the assessment procedure.

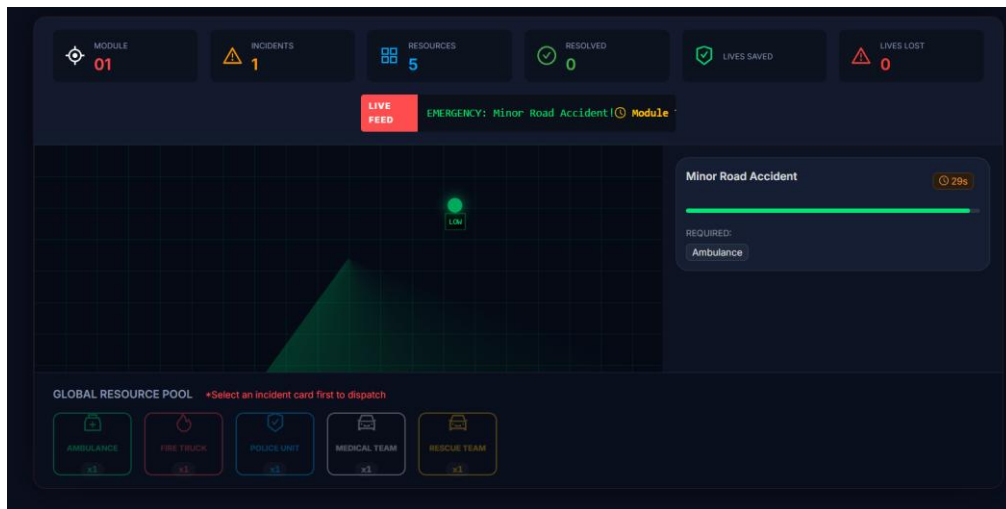


Figure 7.6 Assessment / Command Center

The Command Center is the primary gameplay interface where students act as emergency dispatchers. It displays active incidents, available resources, timers, and system notifications while continuously recording behavioural interactions.

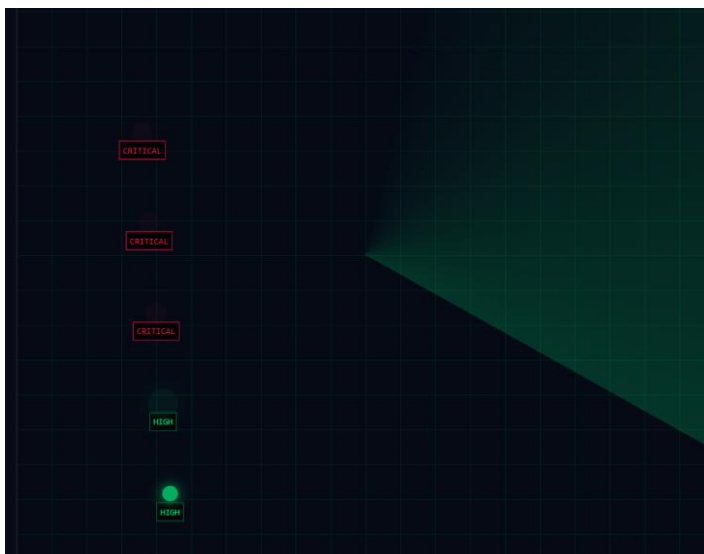


Figure 7.7 Radar Map

The Radar Map visually represents the locations of active emergency incidents within the simulation. Students use this component to monitor emergencies and respond by dispatching the appropriate emergency resources.

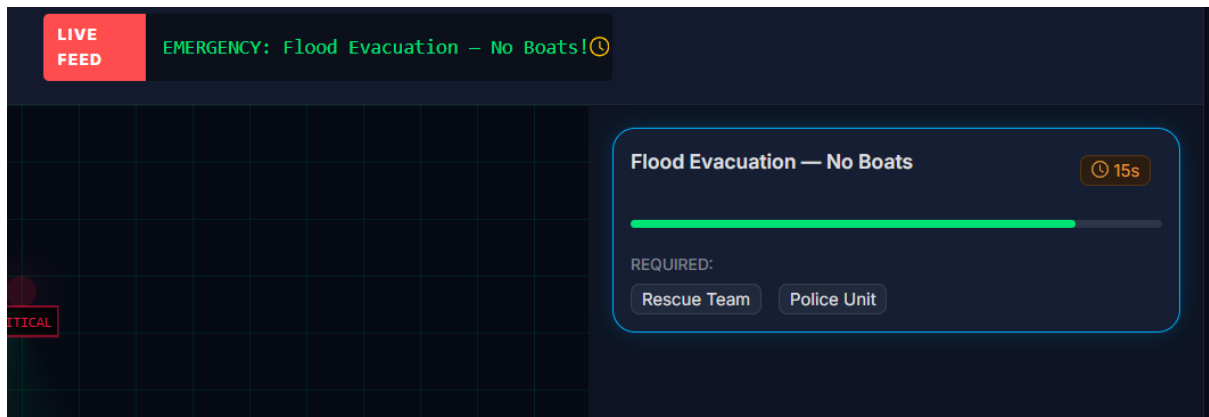


Figure 7.8 Incident Feed

The Incident Feed displays real-time emergency requests with detailed information such as incident type, priority level, and remaining response time. Students must quickly analyse each incident before dispatching resources.



Figure 7.9 Resource Bay

The Resource Bay presents all available emergency response units, including ambulances, fire trucks, police vehicles, and rescue teams. Students allocate these resources strategically to resolve emergency situations.

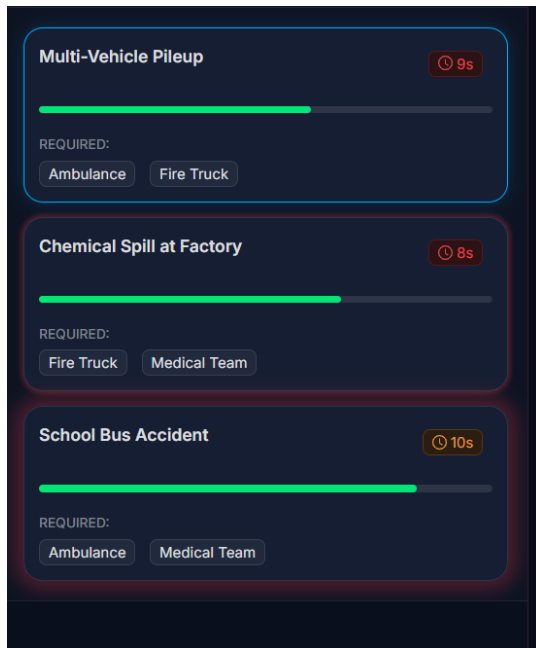


Figure 7.10 Module 2 – Stress Induction

This screen represents the Stress Induction module, where students face multiple simultaneous emergencies, limited resources, and strict time constraints. The module evaluates stress tolerance through behavioural interactions under pressure.

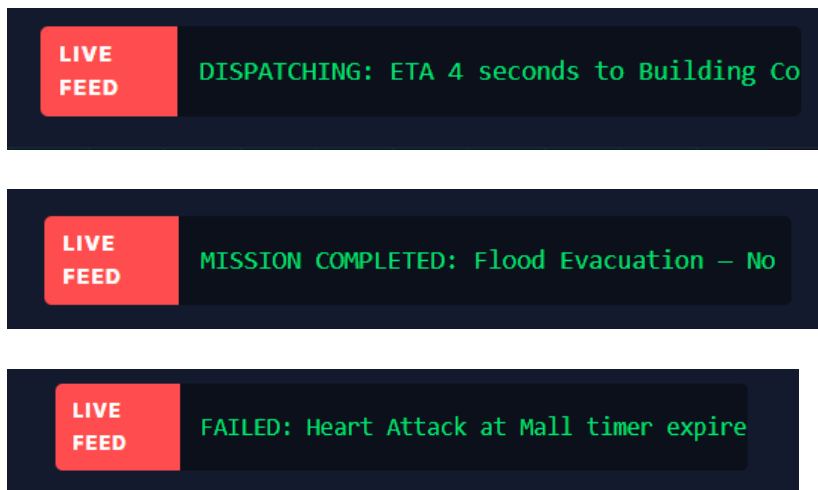


Figure 7.11 Failure & Recovery Module

The Failure and Recovery module introduces unexpected failures such as equipment breakdowns or incorrect information. Students are required to recover quickly, modify their decisions, and continue the assessment successfully.

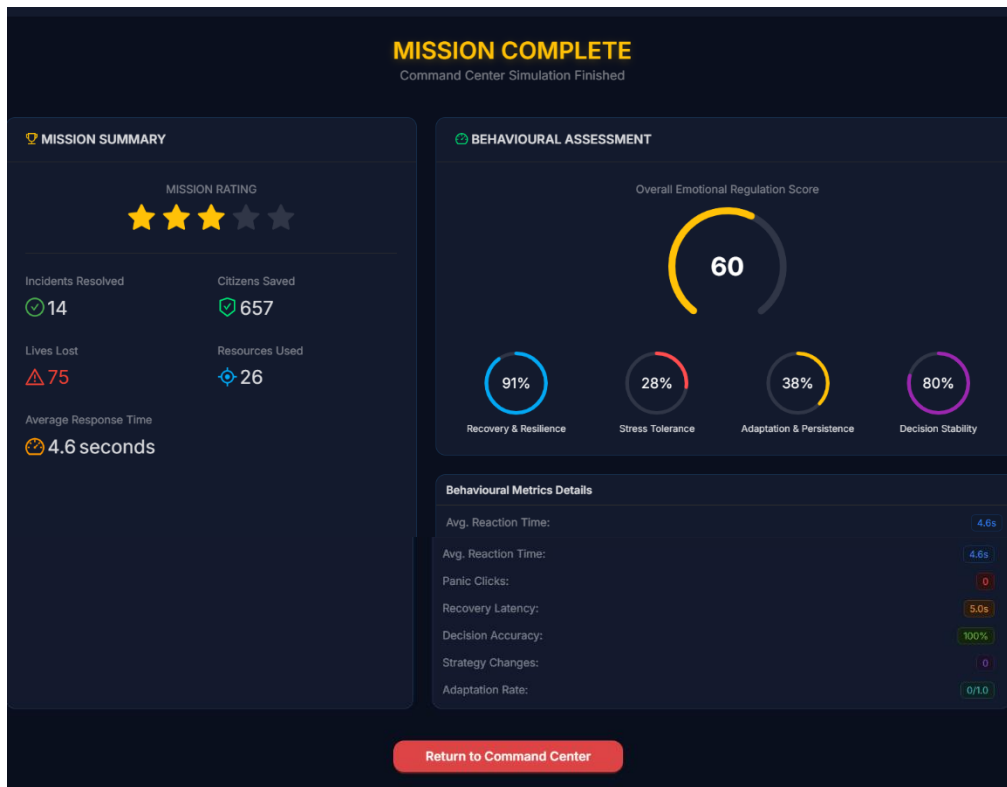


Figure 7.12 Result Page

The Results page displays the overall emotional regulation score, module-wise performance, behavioural metrics, and weighted psychological scores. It provides students with detailed feedback based on their assessment performance.



Figure 7.13 Admin Dashboard

The Admin Dashboard provides administrators with a centralized interface for monitoring student assessments, viewing behavioural scores, analysing performance trends, and managing assessment data.

AnalyticsAssessmentsStudentsSessionsEvent Logs

Recent Assessments

Export CSV

Student	Score	Date	Actions
Student User student@crisisapp.com	60/100	Jul 04, 2026	View Timeline
evlin ev12@gmail.com	62/100	Jul 04, 2026	View Timeline
evlin ev12@gmail.com	60/100	Jul 04, 2026	View Timeline
evlin ev12@gmail.com	38/100	Jul 04, 2026	View Timeline
evlin ev12@gmail.com	55/100	Jul 04, 2026	View Timeline

Figure 7.14 Student Assessment

The Student Assessment page presents detailed information showing details of scores for individual students. Also, can be exported in terms of CSV.

Event Logs

Filter by event typeFilter by module

Timestamp	Student	Event Type	Module	Scenario	Reaction (ms)	Resource	Correct
Jul 04 14:45:19	Student User	INCIDENT_SELECTED	M3	m3_s1	—	—	✗
Jul 04 14:45:18	Student User	SCENARIO_CREATED	M3	m3_s1	—	—	✗
Jul 04 14:45:16	Student User	MODULE_STARTED	M3	—	—	—	✗
Jul 04 14:42:47	Student User	SCENARIO_CREATED	M3	m3_s3	—	—	✗
Jul 04 14:42:46	Student User	SCENARIO_COMPLETED	M3	m3_s2	—	—	✗
Jul 04 14:42:42	Student User	RESOURCE_DISPATCHED	M3	m3_s2	—	—	✗
Jul 04 14:42:42	Student User	RESOURCE_SELECTED	M3	m3_s2	—	Police Unit	✗
Jul 04 14:42:42	Student User	RECOVERY_COMPLETED	M3	m3_s2	—	—	✗
Jul 04 14:42:42	Student User	CORRECT_DECISION	M3	m3_s2	8354ms	—	✗
Jul 04 14:42:38	Student User	RESOURCE_DISPATCHED	M3	m3_s2	—	—	✗
Jul 04 14:42:38	Student User	RESOURCE_SELECTED	M3	m3_s2	—	Rescue Team	✗
Jul 04 14:42:36	Student User	INCIDENT_SELECTED	M3	m3_s2	—	—	✗

Figure 7.15 Event Logs

The Event Logs page displays the chronological sequence of behavioural events generated during gameplay. These logs include user interactions, dispatch decisions, recovery events, adaptive responses, and timestamps, supporting behavioural analysis and assessment evaluation.



Figure 7.16 Student Analytics

The Student Analytics page presents detailed behavioural metrics including reaction time, decision accuracy, recovery latency, stress tolerance, adaptation rate, and overall emotional regulation scores for individual students.

CHAPTER 8

FUTURE WORK

8.1 Introduction

The current version of Crisis Dispatcher successfully implements a simulation-based behavioural assessment platform for evaluating emotional regulation among higher secondary students. The developed MVP includes authentication, emergency response simulation, behavioural event logging, psychological scoring, and administrative analytics. Although the proposed system meets the objectives of the MVP, several enhancements can further improve its functionality, scalability, and research potential.

8.2 Proposed Future Enhancements

The following enhancements are proposed for future versions of the Crisis Dispatcher platform:

1. Artificial Intelligence-Based Behavioural Analysis

Machine learning algorithms can be integrated to analyse behavioural patterns and predict emotional regulation levels more accurately. AI models can identify hidden behavioural trends and provide personalized recommendations based on historical assessment data.

2. Expanded Emergency Scenarios

Additional emergency scenarios such as natural disasters, industrial accidents, medical emergencies, cyber incidents, and large-scale disaster management situations can be incorporated to increase assessment diversity and realism.

3. Mobile Application Support

A mobile version of the Crisis Dispatcher platform can be developed to allow students to participate in assessments using smartphones and tablets, improving accessibility and usability.

4. Advanced Analytics Dashboard

Future versions can include advanced analytical visualizations such as behavioural trend analysis, comparative performance reports, interactive charts, and long-term student progress monitoring.

5. Cloud Deployment

The application can be deployed on cloud platforms to support larger numbers of concurrent users, improve system availability, and simplify maintenance through centralized data management.

6. Institutional Integration

The platform can be integrated with school learning management systems and educational portals to support continuous behavioural assessment and centralized student performance management.

7. Enhanced Gamification

Future versions may introduce additional gamification features such as achievement badges, progressive difficulty levels, dynamic mission generation, and performance rewards to further improve student engagement.

8.3 Summary

The Crisis Dispatcher MVP establishes a strong foundation for simulation-based behavioural assessment using serious game technology. Future enhancements involving artificial intelligence, cloud computing, advanced analytics, and expanded simulation scenarios can further improve the accuracy, scalability, and educational value of the system. These developments have the potential to transform the platform into a comprehensive behavioural assessment solution suitable for educational institutions and research applications.

CHAPTER 9

CONCLUSION

9.1 Conclusion

Emotional regulation plays an important role in enabling individuals to make effective decisions, adapt to changing situations, and recover from stressful events. Conventional assessment techniques primarily depend on self-report questionnaires or clinical observations, which may not accurately reflect behavioural responses in real-time situations. This project addressed this limitation by developing Crisis Dispatcher, a simulation-based serious game that evaluates emotional regulation through objective behavioural analysis.

The developed system provides an interactive emergency response simulation in which higher secondary students (Grades 11–12) assume the role of an emergency dispatcher responsible for managing multiple incidents under varying levels of stress. The assessment is structured into four modules: Baseline Assessment, Stress Induction, Failure and Recovery, and Adaptive Challenge. These modules progressively evaluate behavioural constructs such as stress tolerance, resilience, decision stability, and adaptability while continuously recording user interactions.

The application was successfully implemented using the MERN technology stack, consisting of MongoDB, Express.js, React, and Node.js. The system incorporates secure user authentication, real-time emergency simulation, behavioural event logging, psychological scoring, and an administrative analytics dashboard. Behavioural interactions including reaction time, decision accuracy, recovery latency, panic clicking, strategy adaptation, and resource allocation are continuously recorded and analysed to generate an objective emotional regulation score.

The developed MVP successfully satisfies the proposed objectives by integrating behavioural science principles with serious game technology to create a practical assessment platform. The system demonstrates how simulation-based environments can provide continuous and

objective behavioural assessment while maintaining student engagement through an interactive gaming experience.

Overall, the Crisis Dispatcher platform establishes a strong foundation for future research in digital behavioural assessment. With further enhancements such as artificial intelligence, advanced analytics, cloud deployment, and additional emergency scenarios, the system has the potential to evolve into a comprehensive behavioural assessment tool suitable for educational institutions, psychological research, and training environments.

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